

Consequence of boar edible tissue consumption on urinary profiles of nandrolone metabolites. I. Mass spectrometric detection and quantification of ...

For the first time in the field of steroid residues in humans, demonstration of 19-norandrosterone (19-NA: 3 α -hydroxy-5 α -estrane-17-one) and 19-noretiocholanolone (19-NE: 3 α -hydroxy-5 β -estrane-17-one) excretion in urine subsequent to boar consumption is reported. Three male volunteers agreed to consume 310 g of tissues from the edible parts (meat, liver, heart and kidney) of a boar. The three individuals delivered urine samples before and during 24 h after meal intake. After deconjugation of phase II metabolites, purification and specific derivatisation of target metabolites, the urinary extracts were analysed by mass spectrometry. Identification was carried out using measurements obtained by gas chromatography/high resolution mass spectrometry (GC/HRMS) ($R = 7000$) and liquid chromatography/tandem mass spectrometry (LC/MS/MS) (positive electrospray ionisation (ESI+)). Quantification was realised using a quadrupole mass filter. 19-NA and 19-NE concentrations in urine reached 3.1 to 7.5 $\mu\text{g/L}$ nearby 10 hours after boar tissue consumption. Levels returned to endogenous values 24 hours after. These two steroids are usually exploited to confirm the exogenous administration of 19-nortestosterone (19-NT: 17 β -hydroxyestr-4-en-3-one), especially in the antidoping field. We have thus proved that eating tissues of non-castrated male pork (in which 17 β -nandrolone is present) might induce some false accusations of the abuse of nandrolone in antidoping. Copyright 2000 John Wiley & Sons, Ltd.